

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>
First, we know the power dissipated by the head light bulb is given as 40 watts and the voltage across it is 12 volts. We can use the formula for power in terms of voltage and current: $P = V * I$. We can rearrange this formula to find the current: $I = P / V$. Substituting the given values, we get $I = 40 / 12$.
Now we can calculate the current: $I = 40 / 12 = 3.33$ amps.
Next, we can use Oh m's law to find the resistance: $R = V / I$. Substituting the given voltage of 12 volts and the calculated current of 3.33 amps, we get $R = 12 / 3.33$.
Now we can calculate the resistance: $R = 12 / 3.33 = 3.6$ ohms.
Therefore, the head light bulb will draw a current of 3.33 amps and have a resistance of 3.6 ohms.
</think><answer> E.